

Lab 1: Simple Linear Regression

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1.1 Pythagorean Win Percentage

1.1.1 Data

We are given a dataset of team-seasons from 2017 to 2021. Each row i represents a team-season, and the columns are:

- i : the index of the i^{th} team-season.
- RS_i : the runs scored by the team over the course of the season.
- RA_i : the runs allowed by the team over the course of the season.
- WP_i : the win percentage of the team at the end of the season.

We want to predict end-of-season win percentage from Runs Scored and Runs Allowed. A team's residual from this model measures how much it over- or under-performed relative to this simple specification. Residuals can reflect luck, management, injuries, schedule, player development, and model misspecification.

1.1.2 The Work of Bill James

Bill James, the godfather of sabermetrics (baseball analytics), created the Pythagorean Win Percentage formula.

$$\widehat{WP}^{(\text{Pythag})} = \frac{RS^2}{RS^2 + RA^2}$$

He made it up, and it works surprisingly well! We plot the data from the 2020 season in Figure 1.1.

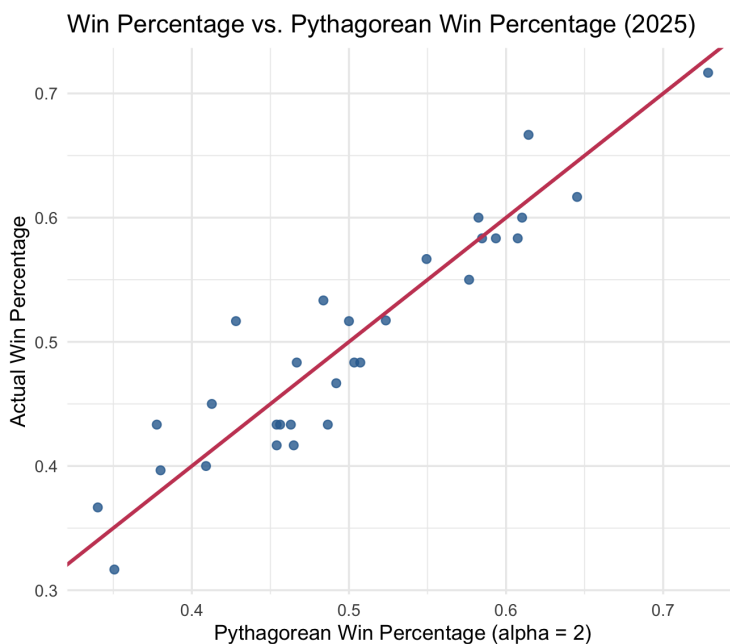


Figure 1.1: Win Percentage vs. Pythagorean Win Percentage for the 2025 season.

However, although the classic Pythagorean exponent (2) is quite good, it is still a modeling choice. You will improve on this by estimating a data-driven exponent with linear regression.

1.1.3 Your Assignment

Task 1: Use linear regression to find an exponent α so that the Pythagorean win percentage

$$\widehat{WP} = \frac{RS^\alpha}{RS^\alpha + RA^\alpha}$$

best fits the data.

Hint: Transform this equation to be **linear** in α :

$$\frac{1 - WP}{WP} = \left(\frac{RA}{RS} \right)^\alpha$$

so that

$$\log \left(\frac{WP}{1 - WP} \right) = \alpha \log \left(\frac{RS}{RA} \right).$$

Then fit a no-intercept regression of $\log \left(\frac{WP}{1 - WP} \right)$ on $\log \left(\frac{RS}{RA} \right)$ (for example, using `~ 0 + log(RS/RA)`). This estimates α on the transformed log-odds scale.

Task 2: Compare your fitted model to Bill James's $\alpha = 2$ model.

- Compute RMSE for each model:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (WP_i - \widehat{WP}_i)^2}.$$

- Report which model has smaller RMSE.
- Create a plot of actual WP versus predicted WP for both models.
- Add the 45-degree line to the plot and use it to compare predictive accuracy.

Task 3: Report uncertainty for your estimate of α .

- Report $\hat{\alpha}$, its standard error, and a 95% confidence interval.
- Interpret the confidence interval in words.
- State whether $\alpha = 2$ is plausible based on your interval.

Task 4: Make a residual diagnostic plot for your fitted Pythagorean model (residuals vs. fitted values).

- What would you expect to see if the linear model is reasonable?
- Do you see curvature, outliers, or changing spread?

1.2 Evaluating MLB General Managers

1.2.1 Data

We are given a dataset of MLB team payrolls and results for each season from 1998-2023. Each row i represents a team-season, and the columns are:

- i : the index of the i^{th} team-season.
- `wp`: the win percentage of the team at the end of the season (denoted WP_i).
- `payroll_median_ratio`: the ratio of the team's payroll to the median payroll of all teams in the league that season (denoted x_i).
- `log_payroll_median_ratio`: $\log(x_i)$.

We want to analyze the relationship between payroll and winning. We will use residuals as a crude measure of performance relative to payroll; this is useful descriptively, but it is not a causal estimate of general manager ability.

1.2.2 Your Assignment

Task 1:

- Remove the 2020 Covid-shortened season.
- Plot `wp` against `payroll_median_ratio`.
- Mark the Oakland A's and the New York Yankees' points on the plot.

- Add the regression line of `wp` as a function of `payroll_median_ratio`.
- Add the regression line of `wp` as a function of `log_payroll_median_ratio`.
- Compare the two model fits on the same figure and explain which one appears to fit better.

Task 2:

- Now for each team-season, calculate the difference between the actual `wp` and the predicted value using `payroll_median_ratio`.
- Do the same with `log_payroll_median_ratio`.
- Add these columns to the dataset.
- Make residual plots (residuals vs. fitted values) for both models.
- Briefly discuss curvature, outliers, or changing spread.
- Find the average difference for each team across all seasons and make two graphs (one for each model) ordered from highest to lowest.
- Change the y-axis scale to wins by multiplying by 162.
- Add a legend to the graphs.

Task 3: Pick one team-season and compute:

- the fitted win percentage;
- a confidence interval for the mean win percentage at that payroll level;
- a prediction interval for an individual team-season at that payroll level.
- State which interval is wider, and explain why.

1.2.3 Important Note:

Let x_i denote `payroll_median_ratio` for team-season i , and consider the following models:

$$\text{Model A: } \widehat{WP}_i = \alpha_0 + \alpha_1 x_i$$

$$\text{Model B: } \widehat{WP}_i = \beta_0 + \beta_1 \log(x_i)$$

We interpret these models as follows:

- Model A: Increasing x_i by a constant value of 1 median payroll adds $\widehat{\alpha}_1$ to $\widehat{WP}_i$.
- Model B: Increasing x_i by $r \times 100\%$ changes $\widehat{WP}_i$ by $\widehat{\beta}_1 \log(1 + r)$.
- For small r , $\log(1 + r) \approx r$, so the change is approximately $\widehat{\beta}_1 r$.

Proof. Let $x'_i = (1 + r)x_i$. Then,

$$\begin{aligned}\widehat{WP}'_i &= \beta_0 + \beta_1 \log(x'_i) \\ &= \beta_0 + \beta_1 \log((1 + r)x_i) \\ &= \beta_0 + \beta_1 (\log(1 + r) + \log(x_i)) \\ &= \widehat{WP}_i + \beta_1 \log(1 + r).\end{aligned}$$

For small r , $\log(1 + r) \approx r$, so the change is approximately $\beta_1 r$. □

Which model do you think is better intuitively? A or B?